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Relative age effects in American professional football

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Abstract: We test for the existence of relative age effects in professional American football. In a sample of 18,898 football players born on or after 1940, there is an excess of January and February births – consistent with a relative age effect associated with calendar year – as well as a slight increase in September births – consistent with the fact that some football players we analyze attended high school in states with fall school cutoff dates. We consider the possibility that relative age effects may affect skilled football positions more than positions relying heavily on player weight, and we find suggestive evidence of this. Lastly, and contrary to what has recently been shown in professional hockey, we find no evidence that misguided preferences for relatively older players lead to selection-based inefficiencies in football player drafting. Our results have implications for evaluating potential football players and speak broadly to the role of physiological factors beyond player control on athletic success.

Keywords: football; performance; relative age; selection.

1 Introduction

Age is an important determinant of success in sports and other domains of human competition. While everyone born in 2010 shares a common birth year, not all individuals born in that year are today the same age and necessarily similar in the sense of physical maturity. To make this point concrete, consider two collections of individuals, one born in January 2010 and a second in December 2010. Because members of the former group are almost a year older than those in the latter, when all 2010-born individuals were still children, we would expect those born in

January 2010 to have been stronger and faster than those born in December. Expressed another way, within the 2010 cohort of individuals who potentially could become professional athletes in a given sport, some members of the cohort are older than others, in some cases by almost a year. This type of observation has motivated research on what are called *relative age effects*.

The importance of relative age across various sports is well-documented, albeit less so regarding football. Addona and Yates (2010) show that there is a plethora of January births and a dearth of December births among North American professional hockey players. Other studies of relative age in hockey are Barnsley and Thompson (1988), Nolan and Howell (2010), and Deaner, Lowen, and Copley (2013). Relative age effects have been found in baseball (Thompson, Barnsley, and Stebelsky 1991), basketball (Delorme and Raspaud 2009), rugby (Till et al. 2010), Australian football (Coutts, Kempton, and Vaeyens 2014), soccer (Musch and Hay 1999; Copley, Schorer, and Baker 2008; Fumarco and Rossi 2018; Rađa et al. 2018), Olympic boxing (Edginton, Gibson, and Connelly 2014), masters swimming and track (Medic et al. 2009), and tennis (Edgar and O'Donoghue 2005). Beyond athletic competition, scholars have found relative age effects among corporate chief executive officers (Du, Gao, and Levi 2012), in elementary school academic performance (Sweetland and De Simone 1987), and surrounding suicide incidence (Thompson, Barnsley, and Dyck 1999).

On the other hand, not all fields of human endeavor are subject to relative age effects. Kniffin and Hanks (2016) find no evidence for such an effect among individuals with doctorates, and Gibbs, Jarvis, and Dufur (2012) argue in favor of a relative age reversal among elite hockey players wherein younger players reap advantages by playing with older competitors. See as well McCarthy, Collins, and Court (2016), who argue in favor of a relative age reversal in rugby and cricket players.

Steingröver et al. (2016) offer two explanations for the existence of relative age effects in sports. According to the *maturation hypothesis*, differences in relative age drive differences between children in size, strength, and coordination. Relatively older players can utilize their physical superiority to outperform relatively younger competitors. Based on the *selection hypothesis*, strong performance in early childhood can increase a relatively older player's

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odds of being chosen to advance to a higher level of competition or increase the likelihood that he or she is selected to join a more intense league or team and through such an arrangement have access to superior training and coaching. Selection-based opportunities can widen a maturation-related talent gap between older members of an age cohort and their relatively younger peers not chosen for such activities.

In American football, which is our focus here, maturation-based and selection-based relative age effects will exist if relatively older youth or middle school players stand out with respect to football ability compared to their relatively young classmates. They may, on account of this, receive more focused coaching or training as a consequence and then use this advantage to become more likely to advance to college football and later the professional ranks.

In a study of professional hockey, Deaner et al. (2013) describe a third mechanism for relative age effects: *selection bias*. They write, “By selection bias, we mean that evaluators (e.g. teachers, coaches) mistakenly grant fewer opportunities (e.g. instruction, access to elite group or team) to relatively younger individuals than is warranted by their latent ability or talent” (p. 1). A relative age effect caused purely by selection bias is not an effect that results from variance across players in maturation or selection in the sense of the two hypotheses described above. Rather, a sport characterized by selection bias-related relative age effects is one in which talent evaluators believe, falsely, that older players perform better or are better risks, all things equal. The result of false beliefs of this nature will be a relative age effect among players that, based on athletic talent alone, should not exist.

The three aforementioned explanations – (a) maturation, (b) selection, and (c) selection bias – for relative age effects are not exclusive. They can all exist simultaneously and, conditional on existing, may interact. Moreover, the three rationales for relative age effects in football may engage different types of football positions. Though Addona and Yates (2010) do not find differences in birth month frequencies across positions in hockey, for now we note that in football certain positions are associated with player weight in excess of 300 pounds and others are associated with refined mechanical skills. As we discuss shortly, these distinctions may be germane to the presence of relative age effects in football.

Our contribution to literature on relative age effects in sport is as follows. After noting the inconsistency in existing findings on relative age in professional American football, we offer an analysis of this subject that incorporates an extensive sample of football players, one larger than

samples used in extant studies. Our collection of players provides strong evidence in favor of relative age effects in football. In addition, our work builds on Beyer et al. (2016)’s suggestion (p. 2969) regarding the role of school cutoff dates in relative age. Namely, we integrate into our empirical analysis of football players’ birth months the idea that differences in the education cutoff dates that delineate one grade from another in a school (i.e. that impose a school-based “birth date” that effectively determines for the purposes of schooling how old a child really is) may drive differences in the relative ages of participants in youth football and hence in the professional ranks. Lastly, we look for evidence of age-based selection bias in American football and do not find any. This suggests that relative age effects in football reflect physical differences across players – based on maturation and selection – as opposed to biased notions that older players are necessarily better.

2 American football and relative age effects in the sport

In this section we provide background on American football and discuss the literature on relative age effects in the sport. We also describe two sources for age effects that drive our empirical analysis.

2.1 Relative age in American football

NFL teams consist of 53 active players, and the league conducts a draft every April. Players enter the NFL both through this draft, consisting of roughly 260 picks, as well as through a post-draft signing period for undrafted free agents.

Existing literature on relative age effects in professional football has mixed findings. MacDonald et al. (2009) offer an analysis of 2144 United States-born players on NFL rosters at the end of October 2004. Assuming a cutoff date of July 31 for participation in youth football, these authors calculate frequencies of births by quarter and find no evidence of a relative age effect. MacDonald et al. hypothesize that the practice in youth football leagues of distributing players across levels of competition based on both age and weight may explain their null finding. As an aside, they note that small cities are overrepresented among the birth locations of NFL players, and see Côté et al. (2006) on birth locations and sports more generally.

In an analysis of 167 players in the American Football Hall of Fame as of 1994, Stanaway and Hines (1995) test

for relative age effects by quarter of birth and find no corresponding evidence. The same is true of Steingröver et al. 2016 study of 2380 NFL players. There is evidence of relative age effects neither in the Canadian Football League (Daniel and Janssen 1987) nor in the X-League (Nakata and Sakamoto 2011), Japan's professional American football league.

On the other hand, Beyer et al. (2016) identify a relative age effect among 2546 American football draftees, finding an excessive number of births between the months of December and February, inclusive; Böheim and Lackner (2012) uncover evidence of a relative age effect in a sample of 1673 football players drafted from 1960 to 2000; and, in a study of college football, Glamser and Marciani (1992) describe birth month patterns in football players attending Western Kentucky University and University of Southern Mississippi in 1989. With 108 and 84 football players, respectively, at these Division I schools, Glamser and Marciani evidence is suggestive albeit not statistically compelling.

2.2 Two sources of age effects in football

Our analysis considers two distinct, potential sources of relative age effects in football. One such source we call a *calendar year effect*. As reviewed above, individuals born in January of a given year are relatively older than those born in December of the same year, and maturation and selection may reward relatively older players.

A second potential source of age effects in football has its origins in the fact that pre-professional football in the United States is often administered through local education systems. Youth football in the United States is managed differently than amateur hockey, soccer, or basketball, sports for which youth travel teams often exist outside of schools. Thus, age effects in football may be confounded by school cutoff dates, i.e. dates that determine a child's school cohort and, to the extent that a football league is affiliated with a school, affect when a child is eligible to play football. In a school with a cutoff date of, say, October 1, the youngest students will be those born in September, and we call an age effect associated with a school cutoff a *school-based age effect*. See Layton et al. (2018) for a study of school cutoff dates and diagnoses of attention deficit-hyperactivity disorder in children.

Addona and Yates (2010) note that youth hockey leagues in Canada have historically used January 1 as the cutoff which determines a player's cohort, and based on the maturation hypothesis this translates into an advantage for children born in January, February, and March. School cutoffs notwithstanding, insofar as many sporting

domains are associated with calendar year age effects, we expect these types of effects to influence the football players we study to the extent that such players pursued athletic success, broadly speaking, when they were young.

We should not expect our two sources of relative age effects, calendar year and school-based, to operate identically across all areas of sport. That said, one factor that differentiates the NFL from other professional sports leagues in the United States is the age at which players can be drafted. To be eligible for the NFL draft, players "must have been out of high school for at least three years and must have used up their college eligibility before the start of the next college football season." Players who are underclassmen or who graduated from college may apply to enter the draft early, but only if they have been out of high school for at least three years.¹ In comparison, sports like hockey and baseball allow players to be drafted much earlier. Casey Mittelstadt, the highest drafted American player in 2017 National Hockey League (NHL) draft, was drafted 8th overall at age 18, just months after his high school graduation.

There are many Mittelstadt-like examples of early draftees across NHL hockey and Major League Baseball. In contrast, the extra years that football players must spend in college provide NFL talent evaluators additional time to identify the most talented individuals and should theoretically provide younger players more time to close any gaps that exist between them and their relatively older peers. In simple terms, the six month advantage a player born in January has over one born in June is greater on a relative basis when players are 18 than when they are 21 or 22. Thus, the NFL's draft system, and therefore the population of NFL players, may be less susceptible to relative age effects compared to a sport like hockey.

3 Data

To address the lack of consensus on relative age effects in football, we draw on an extensive sample of American professional football players extracted from the *Pro Football Reference* (PFR) online database.² This database is also used in Venkataramani, Gandhavadi, and Jena (2018), a study of the relationship between mortality and playing in the NFL. Our individual-level data consist of details on

¹ For more information on the NFL Draft, see <https://operations.nfl.com/the-players/the-nfl-draft/the-rules-of-the-draft> (last accessed March 4, 2018).

² This database is available at ProFootballReference.com, and we scraped this site on March 10, 2018, using OutWit Hub. Scraping code and corresponding data are available from the authors.

18,898 football players whose birth years range from 1940 to 1996. While the PFR database includes 6446 players whose birth years are between 1896 and 1939, we ignore these individuals as they predate the 1970 American Football League (AFL)-National Football League merger by at least a decade. A birth year cutoff of 1940 allows us to incorporate players who would have been 20 years or older at the time of the AFL's formation in 1960 and therefore hypothetically old enough to turn professional soon thereafter. For each player in our purview, we collected name, position, birthday, initial high school attended, regular season games played, and career approximate value, a statistic developed by PFR to judge player value and impact over the course of a career.

Given the range of birth years we consider here, our analysis includes players who competed – or are presently competing – in what we consider to be the modern version of the NFL. Moreover, any player who participated in the AFL, even if the entirety of his career took place in the pre-merger period (prior to 1970), is included in the PFR database. Though some might argue that, strictly speaking, the modern NFL started after the AFL merger in 1970, players participating in latter league prior to the merger were for all practical purposes NFL players and therefore constitute viable observations for a study considering relative age and its impact on reaching the highest level of professional football success.

4 Results

Our results on relative age effects in football are broken into four sections. We first consider calendar age effects among all football players born in 1940 or later. We look for these effects by month and, in an extension of MacDonald et al. (2009), by quarter. Second, we turn to school-based age effects. Third, we consider interactions between relative age and player position. Fourth and finally, we draw on recent research focusing on the NHL draft and look for NFL draft inefficiencies caused by age-based selection bias.

4.1 Calendar year age effects among players born in 1940 or thereafter

Figure 1 displays birth months for the 18,898 football players in our sample who were born in or after 1940. Two features of this figure are notable.

First, the month with the most births is January, the initial month of the year. This is what one would expect if there are calendar year relative age effects in football

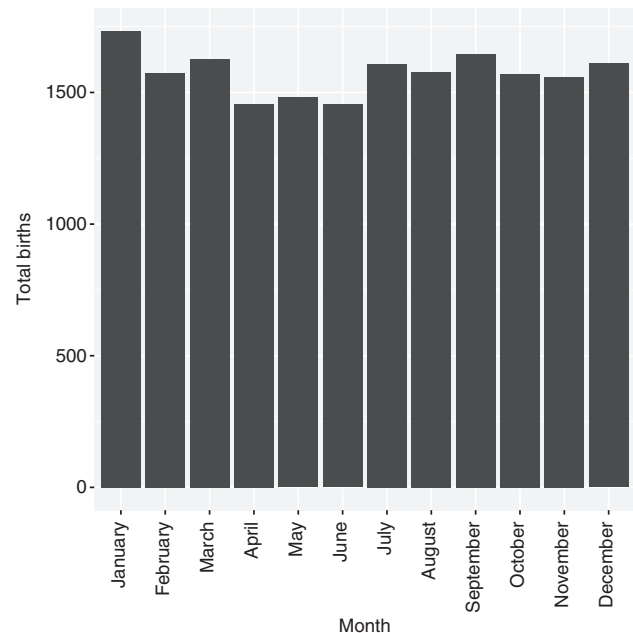


Figure 1: Player birth months.

on account of maturation or selection. Moreover, although February does not appear to have an excessive number of births, March does. Second, the month with the greatest number of births after January is September, and, loosely speaking, the months of July, August, and September seem to have elevated numbers of births. This is roughly consistent with the presence of school-based relative age effects.

If there are calendar year relative age effects in football, then the distribution of birth months shown in Figure 1 should differ from the distribution of birth months for non-football players. With respect to this latter distribution, births in the United States tend not to be uniform across months. This is not entirely surprising, of course, as months vary in how many days they have; this feature could confound the bar heights in Figure 1 and February's bar height in particular. The Centers for Disease Control and Prevention (CDC) publishes data on month-by-month births for the general population in the United States, and we gathered birth data from 2010.³ According to the CDC, there were an average of 334,000 births per month in the United States in 2010, with a standard deviation of 13,850 births. We assume that we can extend 2010 births per month back to 1940.

³ CDC monthly birth counts are available at https://www.cdc.gov/nchs/products/nvsr/monthly_provisional_notice.htm (last accessed February 22, 2018). The 2010 data we use are from the CDC publication titled, "Provisional monthly and 12-month ending number of live births, deaths, and infant deaths and rates: United States, January 2009–December 2010."

Table 1: Monthly births among all football players.

Month	Observed	Expected	Difference
January	1734	1527.68	14%
February	1575	1428.67	10%
March	1628	1598.41	2%
April	1457	1537.11	−5%
May	1481	1551.26	−5%
June	1454	1579.55	−8%
July	1607	1626.70	−1%
August	1579	1659.70	−5%
September	1646	1669.13	−1%
October	1569	1593.69	−2%
November	1558	1537.11	1%
December	1610	1588.98	1%

For our sample of 18,898 football players, Table 1 contains the number of observed births each month, the corresponding expected number of births conditional on CDC birth rates, and percentage differences between these two quantities. The χ^2 statistic for this table is approximately 66.3, which is significant at a conventional test level ($p < 0.001$). We can thus reject the hypothesis that professional football player births follow typical monthly birth rates in the United States.⁴

Per Table 1, January has the largest percentage difference (approximately 14 percent) between observed and expected births, and the second month of the year, February, has the second greatest percentage difference (approximately 10 percent). These statements about January and February are stronger than our earlier comments about bar heights in Figure 1, and this is because the numbers in Table 1 control for varying birth totals per month.

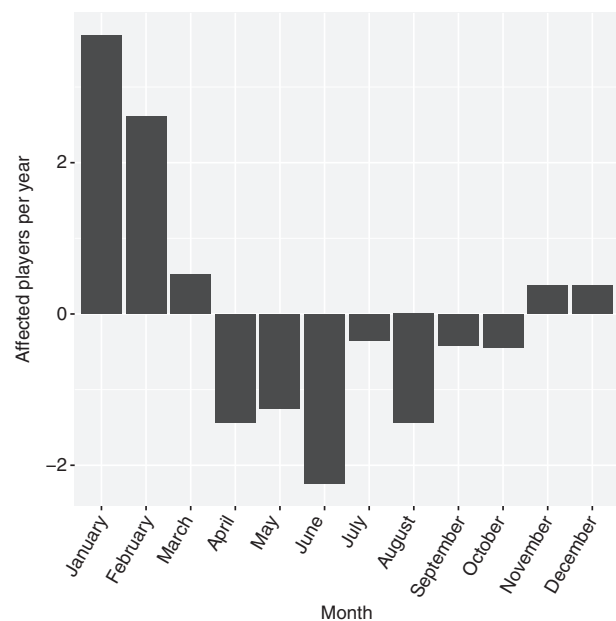
The percentage differences in Table 1 are not monotonically-decreasing across months, and most notably there are positive differences in November and December. Sampling variance notwithstanding, there are a number of potential explanations for the non-monotonicities in the table. One explanation is school-based age effects, and a second explanation is the fact that different football positions may be associated with different types of relative age effects. A third explanation is that football is a sport played mainly in fall and winter; for this reason, being born in late fall/early winter may be beneficial for player development. We consider these explanations shortly.

⁴ The PFR database we use here does not contain data on player socioeconomic status nor on the parents of the players we study. Insofar as birth parent socioeconomic status is correlated with child birth month (e.g. Buckles and Hungerman 2013), in principle we could develop a more-refined null distribution for birth months if we knew more about player families.

One might be concerned that our assumption of constant birth rates per month since 1940 has biased our results on relative age effects. To check this possibility, we gathered daily birth data from the CDC from 1968 (the first year of data availability) through 1996 (the last birth year in our sample of football players), inclusive. According to the CDC, there were 86,403,129 recorded births in the United States in this time period. From these 86,403,129 births, we calculated monthly birth rates and then carried out a chi-squared test of observed birth months for the 10,240 players in our sample born between 1968 and 1996, inclusive, against these rates. This test rejects ($\chi^2 \approx 33.7$, $p \approx 0.0004$), which is consistent with our earlier test result, one that assumed constant monthly birth rates since 1940. As an aside, the mean absolute difference between monthly birth dates in 2010 and those from our aggregated 1968–1996 period is approximately 0.1 percentage points, which implies that American birth rates by month appear not to have changed dramatically since 1968.

Returning to our overall sample, how large are the relative age effects identified in Table 1? The football players we have studied thus far were born across 56 years, and Figure 2 plots the difference between observed and expected counts in Table 1, divided by 56.

Regarding age effects on an annual basis, Figure 2 shows that there are approximately six excess players per year with January through March birth months. Moreover, the number of missing summer births per year among football players is fewer than ten.

**Figure 2:** Per year counts of affected players by month.

To put these two counts in perspective, Addona and Yates (2010) find in their study of relative age in hockey that there are a total of approximately 72 excess players with January through March births among 1184 Canadian NHL players born between 1970 and 1987 (see their Table 4 on p. 7 and note that Addona and Yates sample includes a “small number of players” born after 1987). This is equivalent to approximately four hockey players per year. Over the same 18 year period, Addona and Yates also identify an average of approximately three and one-half missing births among hockey players born from October to December. The 2018 NHL draft featured 217 total picks while the NFL draft typically has 256. Using draft size as a rough measure of the number of players entering the sport professionally, we note that $6/256 > 4/217$.

That said, our NHL-NFL comparison is limited for two reasons. First, the NHL draft supplies players to both the NHL itself and to the American Hockey League and East Coast Hockey League, two professional hockey leagues operating as minor leagues to the NHL. The NFL does not presently have similar minor leagues. Moreover, Addona and Yates study only Canadian born NHL players. Both of these caveats imply that the aforementioned four hockey players per year is an underestimate of the true number of NHL players whose careers are influenced by relative age.

As a robustness check on the number of excessive NFL player births around the month of January, we find similar results to the above if we restrict attention only to players born on or after 1990. See Figure 3. As shown in this figure,

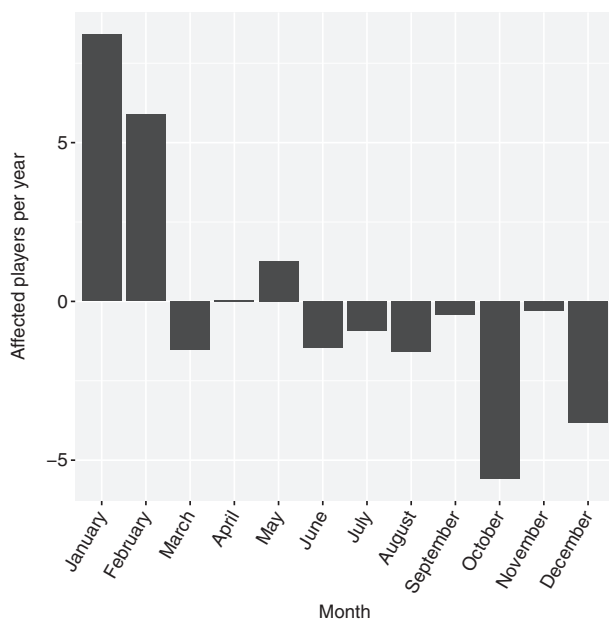


Figure 3: Per year counts of affected players by month, players born on or after 1990.

there are approximately 8.4 excessive January births per year, approximately 5.9 excessive February births per year, and -1.5 excessive March births per year. Thus, our conclusion regarding the overall number of football players affected by relative age is not an idiosyncrasy of considering football players who played in the NFL decades ago.

Of the 18,898 football players whose birthdates are included in Table 1, there are 649 who did not play in any professional football games. If we exclude this set of players from our analysis of births by month, the χ^2 statistic for a modified Table 1 is 63.2, remaining significant at a conventional level ($p < 0.001$). Thus, the aforementioned 649 football players – who perhaps should not be classified as players after all – are not pivotal to our main result on relative age effects in football.

We earlier noted that MacDonald et al. (2009) identify place of birth (large city versus small city, for example), as opposed to time of birth within year, as important in American football player success. In light of MacDonald et al., which considers football players who were on NFL rosters in 2004, we aggregate the football player monthly births depicted in Figure 1 by quarter, matching the approach in MacDonald et al. study. Under this approach, Quarter 1 is defined as August, September, and October; Quarter 2 as November, December, and January; Quarter 3 as February, March, and April; and, Quarter 4 as May, June, and July. This grouping yields Table 2, and with quarterly CDC birth rates as the null distribution for births the χ^2 statistic for this table is approximately 28.3 ($p < 0.001$).

The largest percentage difference (14 percentage points) in Table 2 is in the second quarter, which contains the month of January. The third quarter includes February, and this quarter also has a positive difference (approximately 10 percentage points) in Table 2, albeit one not as large as that pertaining to the second quarter. The table's quarterly result are thus qualitatively consistent with the month-by-month breakdown of football player births depicted in Table 1. Overall, our exercise of aggregating player births by quarter leads us to suspect that the

Table 2: Quarterly births among all football players.

Quarter	Observed	Expected	Difference
1	4794	4922.53	–3%
2	4902	4653.77	5%
3	4660	4564.19	2%
4	4542	4757.51	–5%

Quarters consist of August, September, and October (1); November, December, and January (2); February, March, and April (3); and, May, June, and July (4).

Table 3: Education cutoff date information.

State	Years with data available	State	Years with data available
California	1990	Minnesota	1975, 1990, 2005
Texas	1975, 1990, 2005	Arkansas	1975, 1990, 2005
Florida	1975, 1990, 2005	Massachusetts	None
Ohio	1975, 1990, 2005	Kansas	1975, 1990, 2005
Georgia	1975, 1990, 2005	Kentucky	1975, 1990, 2005
Pennsylvania	None	Iowa	1975, 1990, 2005
Louisiana	2005	Nebraska	1975, 1990, 2005
Illinois	1975, 1990, 2005	Oregon	1975, 1990, 2005
Alabama	1975, 1990, 2005	Connecticut	1975, 1990, 2005
North Carolina	1975, 1990, 2005	Utah	1975, 1990, 2005
New Jersey	None	Hawaii	1975, 2005
Michigan	1975, 1990, 2005	District of Columbia	1990, 2005
Virginia	1990, 2005	Nevada	1975, 1990, 2005
Mississippi	1990, 2005	Idaho	1990, 2005
New York	1975, 1990	West Virginia	1990, 2005
South Carolina	1975, 1990, 2005	Montana	1975, 1990, 2005
Tennessee	1990, 2005	Delaware	1975, 1990, 2005
Washington	1990, 2005	New Mexico	1975, 1990, 2005
Maryland	1990, 2005	South Dakota	1975, 1990, 2005
Missouri	1975, 1990, 2005	New Hampshire	None
Oklahoma	1975, 1990, 2005	Wyoming	1975, 1990, 2005
Arizona	1975, 1990, 2005	North Dakota	1975, 1990, 2005
Indiana	1990, 2005	Rhode Island	1975, 1990, 2005
Wisconsin	1975, 1990, 2005	Maine	1975, 1990, 2005
Colorado	1990	Alaska	1975, 1990, 2005
		Vermont	2005

States ordered by number of players born 1940 or after.

null result on age effects in MacDonald et al. study of football – this analysis draws on a set of 2144 football players – may be due to a lack of statistical power.

We have already noted that MacDonald et al. focuses on players on NFL rosters in a single year, 2004, and this has both advantages and disadvantages. MacDonald et al. do not risk confounding their estimates of relative age effects in 2004 with, say, relative age effects from several decades prior. However, the smaller sample size associated with a single season's worth of football players risks limitations in power.

To explore this tradeoff, we consider a window of football seasons, 2001–2007, centered on 2004. From 2001, we subtract 22 years (most college students graduate when 21 or 22) and then five more years, since the average career length of NFL players has been approximately five years albeit is shrinking.⁵ From 2007, we subtract 22 years. These calculations yield a birth year window of 1974–1985. If we replicate Table 2, restricting attention to players born

between 1974 and 1985, inclusive, we find 4278 players and $p \approx 0.04$. This finding is marginally statistically significant. If we change our window of football seasons to 2002–2006, our number of players drops to 3570 and the p -value for a quarterly χ^2 test based on these players rises to approximately 0.07.

These calculations highlight the tradeoffs inherent in MacDonald et al. and our analysis. While our approach, drawing on a large sample of players, shows strong evidence of relative age effects in football, it does not have leverage over season-specific effects or potential temporal variation in these effects.

4.2 School-based relative age effects

We now turn to our second potential source of relative age effects in football, school cutoff dates. To investigate the potential that these dates are related to performance among American football players, we gathered information on cutoffs used by states. To this end, Colasanti (2007) provides us with school cutoff dates among states in the years 1975, 1990, and 2005. Table 3 summarizes what we know, by state, about these dates. In this table, states are

⁵ See “The Shrinking Shelf Life of NFL Players,” February 29, 2016, available at <https://www.wsj.com/articles/the-shrinking-shelf-life-of-nfl-players-1456694959> (last accessed March 4, 2018).

listed in order of total players analyzed in this section, starting with California and ending with Vermont. For some states (e.g. Texas), we have three snapshots of their school cutoff dates based on Colasanti. For others (e.g. California), we have only one snapshot.

Ignoring players raised in states that allow for local (sub-state) control of school cutoff dates, the 153 players in our sample of 18,898 whose first attended high school was outside the United States, and all players born before 1970 (due to inconsistent information on school cutoff dates), we select from our sample of players those for whom an applicable state cutoff date at time of birth can be logically deduced. This yields a group of 3565 individuals.

Consider a football player from Texas. From Colasanti, we know how Texas school cutoffs evolved from 1975 to 1990 to 2005. Namely, Texas used the same school cutoff date – September 1 – in these three years. We thus include in this section's analysis players born in Texas from 1970 and through 1996. Why do we begin in 1970? A player born in 1970 will have been five years old by 1975, the year that begins the period in Texas in which we know there were consistent school cutoff dates. Though 2005 is later than the birth year (1996) of the youngest players in our overall sample, having school cutoff data from 2005 in conjunction with cutoff data from 1990 informs our ability to infer what school cutoff affected players born between 1990 and 1996.

In the case of New York, Table 3 shows that we have information on this state's school cutoff dates for the years 1975 and 1990. According to Colasanti, New York's cutoffs did not change between 1975 and 1990 but were decided locally in 2005. In other words, at some point between 1990 and 2005, New York began to allow local control of school cutoffs. Accordingly, the only New York players (meaning, players who attended high school in New York), we include in our analysis here are those born between 1970 and 1985, that is, prior to the first possible change for New York's school cutoffs. The rationale for 1970 here is identical to that noted in the previous Texas example.

We assume that a player's state is determined by the first high school that he attended, as listed in the PFR database. We break known school entrance dates into the categories "Fall" and "Winter." The former category includes cutoff dates in August, September, October, and November and the latter category, cutoffs in December, January, and February.

Students moving across states during high school, from fall cutoff states to winter cutoff states, could in principle confound our analysis. However, if for strategic reasons a family with a son born in August were to move, before that son entered high school, from a non-fall cutoff

state to a state with a September 1 cutoff, we would classify this student as having gone to school entirely in a fall cutoff state. To the extent that the student prospers on account of his move, this will make it harder for us to detect school-based relative age effects.

Figure 4 displays patterns in birth months for players in fall and winter cutoff states, and there are many more of the former (3147) than the latter (418). With attention on the former group, we find an excessive number of births early in the calendar year and in the late fall. This is consistent with the presence of both calendar year and school-based relative age effects.

Table 4 reports birth months for 3147 births that took place in fall-cutoff states. Looking at the "Difference" column in the table, there is a positive January spike, which is consistent with Figure 4, a slightly smaller February spike, and then a still smaller March spike. Percentage differences then decrease until September, when an upward trend appears. The χ^2 statistic for Table 4 is approximately 21 ($p = 0.034$), and, overall, this table contains evidence of both calendar year and school-based relative age effects.

We have noted that football across the United States is a fall-winter sport and, accordingly, that being born in the fall may be beneficial for player development. Because school years and the start of football seasons are roughly coincidental, we cannot distinguish between fall-winter births that are valuable for a player on account of a school cutoff versus those that are valuable due to the timing of football. This is simply a matter of two mechanisms for relative age effects that work in similar directions. In order to investigate whether sport timing is important in football player development, we would need to find a professional football environment in which the sport were played in, say, spring-summer. This research design matter is beyond our present purview, and in the conclusion we return to the subject of design.

Of our football players with fall birthdays, some attended religious high schools which, by virtue of their private nature, may have had different cutoff dates than public schools (or, possibly, no formal cutoff dates at all). With this in mind, we search in our PFR database for players who attended Catholic high schools by matching in a case-insensitive way the following terms against high school names in our dataset of football players: Cathedral, Catholic, Christian, Sacred, Saint, and St. Of the players who attended high school in fall cutoff states and were born after 1970, our name-matching approach identifies 99 who attended a religious school.

When we remove these 99 individuals from our analysis, our results are qualitatively the same as before, with a χ^2 statistic of 20.8 ($p = 0.036$). This implies that our

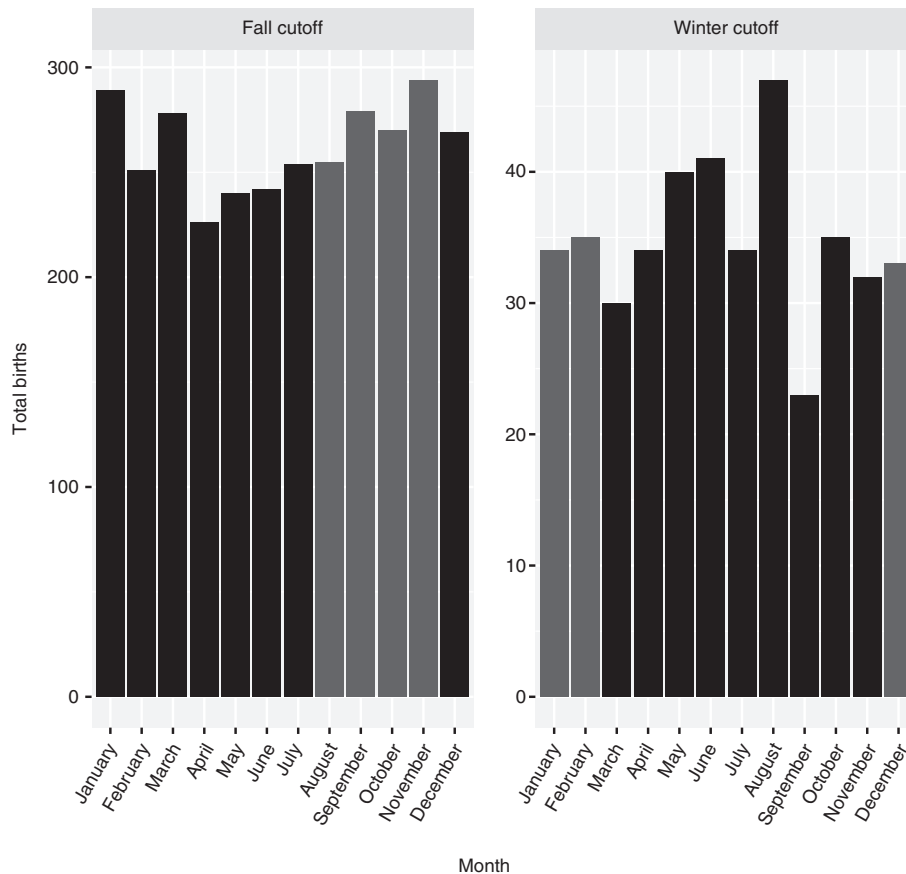


Figure 4: Player birth months across fall and winter cutoffs. Gray bars denote months in a given cutoff period.

Table 4: Monthly births among football players in fall cutoff states.

Month	Observed	Expected	Difference
January	289	254.40	14%
February	251	237.91	6%
March	278	266.18	4%
April	226	255.97	−12%
May	240	258.32	−7%
June	242	263.04	−8%
July	254	270.89	−6%
August	255	276.38	−8%
September	279	277.95	0%
October	270	265.39	2%
November	294	255.97	15%
December	269	264.61	2%

findings about calendar year and school-based relative age effects for football players in fall cutoff states hold even after removing players who lived in such states but attended religious schools, which might have used more flexible cutoffs.

Our sample of football players born in winter cutoff states has only 418 individuals. This small number will diminish our ability to detect any winter birth spikes,

if they exist, and indeed we do not find a statistically significant difference between birth months for these players and birth months for non-winter players ($p = 0.39$).

Overall, our analysis in this section provides evidence of school-based relative age effects among football players. The evidence illustrates how institutional rules relating to an adolescent's educational and life course can interact with physiological development.

4.3 Relative age effects and position

Here we investigate position-specific relative age effects by grouping players born after 1940 into one of three position categories, *Skill*, *Combination*, and *Line*. We define skill positions as quarterback, running back, specialist (includes kickers, long snappers, and punters), wide receiver, and variations thereof. Combination positions include all variations of fullback, linebacker, and tight end. Line positions include all variations of offensive and defensive lineman.

Anecdotally, we know that our position groupings are used on Dartmouth College's football team and others to

classify football players based on the ratio of speed to size needed for a given position. Skill positions generally require running speed and agility and are responsible for throwing the football, running with it, preventing others from catching it, or kicking it. In contrast, line positions are less speed and skill-based. Players on the offensive side of the line, like center, guard, and tackle, are responsible for blocking defensive players; those on the defensive line, like defensive end, defensive tackle, and nose tackle, are responsible for rushing the quarterback and stopping running plays. Though there are unique techniques involved in these positions, size is paramount. NFL offensive and defensive lineman often weigh at least 300 pounds.⁶ Finally, combination positions exist in a space between skill and line groups, both literally on the field and figuratively in terms of desired speed to size ratio.

Why might relative age effects in football vary by position? Line positions in particular generally require that a player engage in close physical contact with another individual, and the skills associated with line play cannot be practiced repeatedly in the way that a skill like throwing or catching a ball can be methodically practiced. Because of their physically demanding nature, line techniques face a limit on how frequently they can be practiced. Moreover, a 2017 *Washington Post* story on football notes that lineman entering college football suffer because of the lack of a youth system that allows them to practice their contact-intensive techniques.⁷ Finally, success in line positions requires utilizing physical strength and weight that, by definition, do not fully develop in players until they are grown.

As to our claim that what we call skill positions in football are in fact more skill-based than line positions,

we note that there is an academic literature on overhead throwing. This literature describes the difficulty of mastering this action, something that is very important for football quarterbacks. For example, see Wilk et al. (2000), Hirashima, Kudo, and Ohtsuki (2003), and Ghorbani and Bund (2017), which cover both baseball and football. We have been unable to find a comparable literature that engages football line techniques like blocking and tackling.

All told, the differences in line and skill positions that we have described here lead us to conjecture that football positions requiring extreme weight and strength may be, all things equal, less prone to relative age effects than more skill-based positions, namely those that involve throwing and catching. Combination positions lie in between line and skill, and we would expect them to be associated with moderate age effects, if any at all.

Parallel to the way we presented earlier results, Table 5 reports by our three position categories – skill, combination, and line – birth months and deviations from expected births. The deviations are, as before, based on CDC birth frequencies per month.

We find evidence of relative age effects for skill and combination players ($p < 0.001$ and $p < 0.05$, respectively), and evidence for the former group is strong. However, our χ^2 test rejection for combination players is to some extent driven by drops (approximately 10.9% and 11.1%) in April and May births, which are not consistent with the theories for relative age effects we have described here. We do not find statistically significant evidence of relative age effects for line positions ($p = 0.13$).

These three tests constitute suggestive evidence that football positions relying more on skill than on weight may be more vulnerable to relative age effects. However, if we carry out a χ^2 test of independence of rows and columns for the 12×3 table of monthly births by position, we do not reject ($p \approx 0.46$). This single test is more compelling than the three separate tests, one per position group. Thus, notwithstanding our stated logic regarding an association between player position type and relative age effects, we cannot conclude from our data that such an association exists.

Given the reliance of some football positions on weight as opposed to pure skill, the possibility of a relationship between position type (in our case, line, or combination) and the presence of relative age effects may be uniquely compelling in football compared to other major sports. Even so, we encourage other scholars interested in relative age effects in sports to explore the issue of skill versus weight and the extent to which requirements of certain positions in sports may be correlated, or not, with youth

⁶ For a discussion of the evolution of player weight in the NFL, see “N.F.L. Linemen Tip the Scales,” *The New York Times*, January 28, 2011, available at <http://www.nytimes.com/2011/01/29/sports/football/29weight.html> (last accessed March 12, 2018). This article reports that 532 participants in pre-season NFL training campus weighed at least 300 pounds. An article on a recent *decrease* in player weight notes that, “[T]he [Atlanta] Falcons’ average weight of about 304 pounds is roughly 10 pounds lighter than the league median and 20 pounds lighter than Washington’s, the league’s heaviest group.” See “N.F.L. Offenses Are Changing, So Some Offensive Linemen Are Shrinking,” *The New York Times*, October 29, 2016, <https://www.nytimes.com/2016/10/30/sports/football/nfl-offenses-are-changing-so-some-offensive-linemen-are-shrinking.html> (last accessed March 12, 2018).

⁷ See “The NFL has an offensive line crisis,” September 13, 2017, available at https://www.washingtonpost.com/news/sports/wp/2017/09/13/the-nfl-has-an-offensive-line-crisis/?utm_term=.fe80ab1aa93f (last accessed March 9, 2018).

Table 5: Monthly births among football position categories.

Month	Skill			Combination			Line		
	Obs.	Exp.	Diff.	Obs.	Exp.	Diff.	Obs.	Exp.	Diff.
January	844	731	15.5%	364	318	14.5%	492	451	9.1%
February	778	683	13.9%	309	298	3.7%	450	422	6.6%
March	766	764	0.3%	351	333	5.4%	488	472	3.4%
April	707	735	−3.8%	285	320	−10.9%	439	454	−3.3%
May	730	742	−1.6%	287	323	−11.1%	445	458	−2.8%
June	671	755	−11.1%	311	329	−5.5%	447	467	−4.3%
July	745	778	−4.2%	358	339	5.6%	480	480	0%
August	777	794	−2.1%	328	346	−5.2%	443	490	−9.6%
September	762	798	−4.5%	346	348	−0.6%	509	493	3.2%
October	741	762	−2.8%	349	332	5.1%	453	471	−3.8%
November	720	735	−2%	317	320	−0.9%	485	454	6.8%
December	796	760	4.7%	333	331	0.6%	451	469	−3.8%

Obs. denotes observed counts, Exp. expected counts, and Diff. differences.

talent. Our evidence on skill players in football is sufficient for us to raise this matter as an intriguing research area for the future.

4.4 Checking for age-related selection bias in the NFL draft

Earlier we noted Deaner et al.'s (2013) argument that talent evaluation in the NHL is systematically biased by age. In particular, these scholars claim that NHL teams systematically undervalue relatively younger players, and their conclusion is based on statistical evidence that such players outperform their draft positions, all things equal. Deaner et al. conclude that NHL teams deliberately – and mistakenly – select relatively older players in the annual hockey draft, thus inducing a relative age effect.

Here we assess whether there is evidence of a similar selection bias in the NFL draft. To address this possibility, we consider a set of 2513 NFL players who were drafted between the years of 2008 and 2017, inclusive. Each player has a draft number where low numbers indicate earlier, and hence better, selections. Since the number of players chosen in NFL drafts fluctuates, we normalize each pick number by dividing it by the number of individuals drafted in its year.

The reasons that we focus here on players drafted between 2008 and 2017 are three-fold. First, the NFL drafts in this period all had seven rounds. Even so, there was slight variation across years in terms of numbers of players drafted, i.e. 253 selections in 2012, 2016, and 2017 and 256 in 2009, 2013, and 2014. Second, during the 2008–2017 time frame, the number (32) of teams in the NFL was constant. Third, the PFR player database on which we rely

contains complete data on almost all (99.4 percent) drafted players between 2008 and 2017. For players drafted prior to 2008, the database sometimes lacks birthdate information on players who were drafted yet never made the roster of a professional football team or simply has no records at all of these types of individuals. This is not a problem for our prior analyses because earlier we deliberately conditioned on professional football players. Here though, where we are looking for evidence of selection bias, it is important that our set of scrutinized players includes drafted players who were ultimately not successful in becoming professional.

How can we know whether relatively younger drafted NFL players on average outperform their (normalized) draft positions? To answer this question we first require a measure of football player success. Armed with this measure, we then need to consider whether, holding draft position fixed, relatively young players had more success in the NFL than relatively old players, all things equal.

In terms of measuring football player success, this matter is confounded by the team aspect of the sport (e.g. Gerrard 2007). In addition, the dimensions of success for a given football player depend on his position. For example, running back effectiveness is typically measured in yards gained while quarterback effectiveness is based on pass completions. To keep our approach as general as possible, our consideration here of selection bias in the NFL draft relies on a measure of player success called “approximate value.”⁸ We briefly mentioned this measure (hereinafter, *Av*) in the introduction, and its key feature is that it allows the contributions of football

⁸ Approximate value is described at <https://www.pro-football-reference.com/blog/index37a8.html> (last accessed August 20, 2018).

players to be compared to another. See Schuckers (2011) for a discussion of Av and other player performance metrics.

Av is confounded by the length of a player's career in the sense that a player accumulates Av points, so to speak, the longer he plays in the NFL. We thus expect older players will have greater Av numbers, all things equal, compared to younger players. This feature of Av leads us to consider Av per season played, but normalizing in this way raises another issue. Consider an excellent football player (A) who played one season and then left the NFL, and consider a second excellent player (B) who played five strong seasons in the NFL. Player B will have a greater Av score than Player A, which captures B's longer career and greater overall productivity. Players A and B, however, will have similar Av per season, which accurately captures their per-season productivity rates. From the perspective of analyzing whether there is selection bias in the NFL draft, it is not obvious that total player productivity is inherently more (or less) meaningful than per season productivity, and hence in this section we consider both Av and Av per season.

Beyond these two measures of player value, we also analyze total games played during a player's career. This measure appears in Deaner et al.'s study of the NHL draft. Our use of both Av (total and per season) and total games played allows us to build directly on Deaner et al.'s analysis of hockey and extend their study with a football-specific metric.

Table 6 describes the positions considered in this section, the rates at which they appear in our analysis of selection bias in the NFL draft, and average games played by position. We see from the table that the most common such positions are defensive lineman (defensive end, defensive tackle, and nose tackle), offensive line (center, guard, and tackle), defensive back (cornerback and safety), linebacker, and wide receiver while the least common positions are quarterback, tight end,

running back, and specialist (kicker, punter, and long snapper).

The reason that some football positions in our data occur more than others is a function of the way that players are used on the field and hence appear on active team rosters. For instance, a team on defense typically plays at least four defensive backs, either three or four defensive lineman, and at least three linebackers at any given time – but, when on offense, only one quarterback, one or two tight ends, and one running back play at a time. A team also typically plays five offensive lineman and between two and five receivers at a time.

Before we present regression results, Figure 5 displays the relationship between Av and normalized draft pick. Each point in the figure denotes a player, the color of each dot connoting whether a player was born in the first three months of the year or not (we drop the two players in our PFR database that are reported as having negative Av). Figure 5 also contains two loess smoothers, which summarize the points in the figure.

Three features of Figure 5 are notable. First, its two smoothers suggest that the relationship between a player's normalized draft number and his subsequent performance in the NFL is not linear. Second, both aforementioned smoothers trend down in normalized draft pick, indicating that, intuitively, higher draft numbers are associated with lower player performance. And third, Figure 5 implies that calendar-based relative age effects may interact with draft choice. Note that part of the light gray line (birthdays

Table 6: Distribution of player positions.

Position	Count	Average Games
Fullback	19	36
Specialist	34	54
Quarterback	94	29
Tight end	134	47
Running back	201	41
Wide receiver	295	41
Linebacker	307	41
Offensive lineman	376	43
Defensive lineman	408	46
Defensive back	484	43

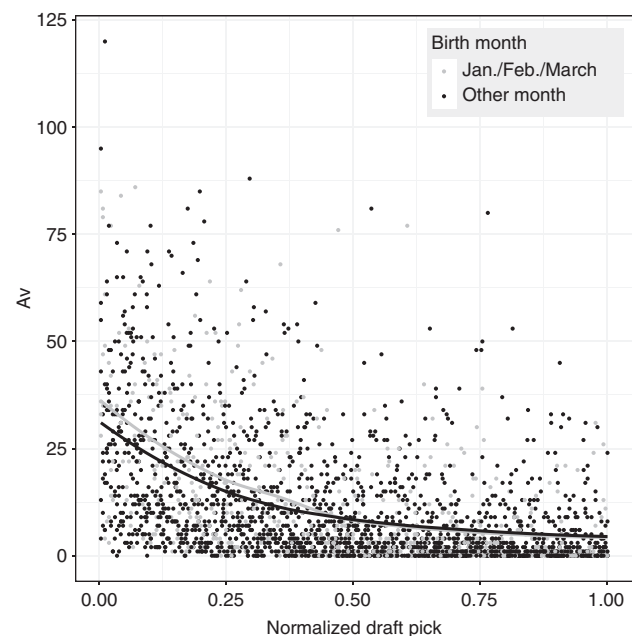


Figure 5: Av and draft pick.

in January, February, and March) in Figure 5 lies above the black line (birthdays in latter months). Had Figure 5 depicted the relationship between player draft number and Av / season, we would observe patterns almost identical to those in the existing figure (additional figure available from the authors).

We now present regression results that explain our three measures of football player success, Av, Av per season, and total games played. We regress each of these measures on two key independent variables: the number of days after January 1 that a player was born (this number could be zero) and a player's normalized draft number. Our regressions also include position fixed effects and, with the exception of the Av per season regression, draft year fixed effects.

For each of our two dependent variables that measure player success, we consider four separate configurations of the key independent variables. In the simplest configuration of these independent variables, both variables (days after January 1 that a player was born and normalized draft number) are linear predictors of player success. In the most complex configuration, both independent variables are permitted to have non-linear effects on player success, and we model these effects with natural splines (five degrees of freedom). To round out our four regressions per dependent variable, we allow for one linear and one non-linear independent variable. Because we have two independent variables, there are two such mixed cases per dependent variable.

For our three regression models of player success, we thus have four regressions each. From each set of four we choose the model with the lowest Akaike information criterion, and for all of our dependent variables this model is the one in which days born after January 1 is linear and normalized draft pick is non-linear. Regression results are in Table 7.

Key in Table 7 is the first row, that corresponding to "Days born after January 1." This row helps us understand whether relative age is associated with football player success, holding draft position constant.

In the Av column of Table 7, the coefficient estimate for "Days born after January 1" is negative and significant at a conventional test level ($p \approx 0.04$). Since relatively younger football players in a year cohort were born long after January 1, it follows that younger players have less football success (as measured by Av) than older players, controlling for draft status. This is not evidence of selection bias along the lines of Deaner et al.'s analysis of the NHL draft. Rather, the significantly negative estimate associated with "Days born after January 1" is evidence of the type of relative age effect we have documented

Table 7: Regression analyses of player performance.

	Dependent variable		
	Av (1)	Av per season (2)	Games played (3)
Days born after January 1	-0.005** (0.002)	-0.001 (0.001)	-0.011* (0.006)
Draft pick, spline term 1	-20.678*** (1.399)	-2.569*** (0.300)	-26.479*** (3.258)
Draft pick, spline term 2	-26.653*** (1.804)	-4.161*** (0.399)	-35.896*** (4.207)
Draft pick, spline term 3	-21.494*** (1.411)	-3.184*** (0.361)	-37.124*** (3.302)
Draft pick, spline term 4	-43.064*** (2.841)	-6.765*** (0.602)	-61.082*** (6.618)
Draft pick, spline term 5	-19.658*** (1.253)	-2.981*** (0.340)	-40.606*** (2.944)
Position: Defensive back	-1.979 (1.389)	-0.720** (0.330)	23.345*** (3.273)
Position: Defensive lineman	-0.520 (1.404)	-0.465 (0.329)	23.309*** (3.308)
Position: Fullback	-3.141 (3.095)	-1.449** (0.698)	32.659*** (7.255)
Position: Linebacker	0.235 (1.451)	-0.068 (0.341)	21.142*** (3.419)
Position: Offensive lineman	2.689* (1.415)	0.008 (0.330)	20.508*** (3.336)
Position: Running back	0.475 (1.540)	-0.135 (0.374)	20.545*** (3.623)
Position: Specialist	0.532 (2.470)	-0.238 (0.507)	49.316*** (5.771)
Position: Tight end	-4.044** (1.659)	-1.096*** (0.385)	25.674*** (3.903)
Position: Wide receiver	-1.263 (1.456)	-0.362 (0.341)	18.738*** (3.428)
Year drafted: 2009	-1.374 (1.137)		-2.369 (2.646)
Year drafted: 2010	-0.786 (1.126)		-5.307** (2.622)
Year drafted: 2011	-2.218** (1.127)		-10.889*** (2.626)
Year drafted: 2012	-3.409*** (1.115)		-16.690*** (2.603)
Year drafted: 2013	-6.263*** (1.123)		-21.691*** (2.619)
Year drafted: 2014	-7.693*** (1.114)		-29.618*** (2.608)
Year drafted: 2015	-10.156*** (1.142)		-34.468*** (2.657)
Year drafted: 2016	-12.294*** (1.140)		-39.223*** (2.653)
Year drafted: 2017	-15.331*** (1.166)		-46.015*** (2.712)
Constant	39.939*** (1.802)	6.223*** (0.337)	73.179*** (4.214)
Observations	2352	805	2352
R ²	0.367	0.380	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Omitted categories are quarterback (among Positions) and 2008 (Year drafted).

previously: relatively younger players have less football success than relatively older players, holding draft position constant.

In terms of other estimates in the first row of Table 7, none speak to relative age in particular and thus are ancillary to our interest in potential age-based selection bias in the NFL draft. We note, though, that “Year drafted” estimates in the first row of Table 7 are decreasing, and this reflects the fact, as noted earlier, that a player’s overall value in his football career is by construction lower, all things equal, when a player is younger.

As an aside, quarterbacks have unusual player productivity patterns that transcend the position indicator variables included in the left column regression in Table 7. Based on this model, we calculated values of Cook’s Distance for each observation in the regression. Approximately 17 percent of the quarterbacks in our sample (total of 94) have elevated leverage, defined here as a value of Cook’s Distance greater than four divided by sample size. The next greatest percentage is slightly under six percent, and this figure corresponds to wide receivers. Given the unique role that quarterbacks play on football teams, we would not want our results in Table 7 to reflect something idiosyncratic about this position. If we re-estimate the regression in the leftmost column of Table 7, this time dropping quarterbacks, our results are qualitatively identical, indicating that the inference we have drawn, above, do not reflect something unusual about quarterbacks.

The middle column (“Av per season”) of Table 7 describes regression results that model player performance normalized by total seasons played (observations are weighted by total seasons). As before, the key variable (“Days born after January 1”) in this set of results is in the first row of the table. While the associated estimate is negative (0.0004), it is not significant at a conventional level ($p \approx 0.214$). This null result, which holds if quarterbacks are dropped from the Av per season regression, is not consistent with the presence of age-based selection bias in the NFL draft.

The third column in Table 7 (“Games played”) describes a censored regression model analyzing total regular season games played. Games played per player is censored from the left at zero since players, by definition, cannot play fewer than zero games. Moreover, games played per player is censored from the right since a player cannot play more regular season games than 16 per year. So, for example, a player drafted in 2017 can have played a maximum of 16 games total, a player drafted in 2016 can have played 32 games total, and so forth. Among our

2513 draftees between 2008 and 2017, there are 203 who played zero games and thus are modeled as left censored; in contrast, 98 are right censored.

There is no evidence in the Games played column of Table 7 that there is selection bias in the NFL draft. Parallel to our Av discussion above, the Games played estimate associated with days born after January 1 is negative, indicating that younger players play fewer games than older players, all things equal. The significance level of the “Days born after January 1” estimate is at best marginal. Regardless, were there selection effects in the NFL draft akin to what Deaner et al. found in the NHL draft, we would expect a positive and significant estimate associated with “Days born after January 1.” Draft year fixed effects associated with total games played are decreasing in time, and this reflects the fact that older players have had the chance to play in more games than younger players, all things equal.

With respect to positions, recall that quarterback is the base category for the total games censored regression in Table 7. That said, the games played column of the table indicates that drafted quarterbacks tend to play disproportionately few games, conditional on being drafted and making an NFL roster. This is evident because the other games played position estimates in Table 7 are positive and statistically significant at conventional levels (p -values vary for the position estimates but are less than 0.01). This presumably reflects the fact that a football team’s quarterback is the most important position on the team and that many drafted quarterbacks are, even after being drafted, not deemed good enough to have long careers. With the exception of the specialist estimate, the various position estimates in Table 7 are not appreciably different from one another. Specialists play many more games than non-specialists.

To summarize, if there were age-related selection bias in the NFL draft consistent with recent findings surrounding the NHL draft, we would expect to see positive coefficient estimates in Table 7 associated with days born after January 1. In fact, we see negative estimates, one of which is statistically significant. These estimates are not only inconsistent with draft-based selection bias but actually suggest that the magnitude of relative age effects in professional football may be unappreciated by the talent evaluators who choose players in the NFL draft. Since our models in Table 7 uncover no evidence of age-based selection bias in football player drafting, this leaves the maturation and selection hypotheses as the logical explanations for the relative age effects in football that we have documented.

5 Discussion

We have shown with a collection of 18,898 American football players that professional success in football is associated with relative age. In particular, successful football players are disproportionately born in January and February and, to a lesser extent, in March. While previous studies have found mixed evidence of relative age effects in American football, entrants in the literature on this subject have not used samples with the breadth of ours. Our results add professional American football to the domains, both sporting and non-sporting, in which relative age affects success.

A notable aspect of football is the way in which a team's players have differentiated roles. There is in this sport a skill versus weight dimension among players that may not mirror position differences in, for example, hockey and basketball, sports in which puck and ball contact, respectively, are regular occurrences for all players on a team. We have incorporated positions in our analysis of age effects in football, and we find suggestive, but not definitive, evidence of such effects in positions that demand speed and skill. We believe that future analysis of relative age effects in sports would benefit by taking a skill versus weight dimension into consideration.

Following a suggestion in Beyer et al. (2016), we uncover evidence of school timing effects based on an analysis of football players born in states that employ a fall academic cutoff date. Our consideration of fall is justified by the structure of youth football competition in the United States and the track by which players proceed to the professional ranks of this sport. Unlike in hockey or basketball, where systems of elite junior travel teams tend to dominate high level amateur competition, football runs almost entirely through local education systems. Players who are successful at their high school programs proceed to college football, with the most talented continuing from there to the NFL. It is true that some communities have youth football programs, Pop Warner for instance, that function outside local education systems. However, these programs generally exist only as an option for players younger than high school age, and they fade in importance as players grow older. Additionally, these youth leagues may use a cutoff date similar to local school districts. Pop Warner's July 31 cutoff date, for example, is not overly dissimilar to a fall academic cutoff such as August or September 1.

Further studies might want to build on Glamser and Marciani (1992) to confirm whether school-based relative age effects impact participation in college athletics more broadly. In the case of football, where college participation

is effectively the only pipeline to the professional ranks, school based relative age effects on college participation would likely persist as players aged towards higher levels of play.

Finally, we extend to the NFL Deaner et al.'s (2013) study of the National Hockey League, which shows that hockey teams systematically undervalue relatively younger players in the annual NHL draft. This form of selection bias can accentuate relative age effects caused by player maturation as well as create relative age effects that, based solely on player physiological characteristics, do not exist.

We find no evidence of age-based selection bias in the NFL draft, and our draft results can be interpreted as implying that talent evaluators in the NFL understand, either implicitly or explicitly, the magnitude of relative age effects in the sport. Why then would age-related selection bias affect the NHL but not the NFL? More generally, what distinguishes sports for which age-based selection bias is an issue from those where it is not? While we can only speculate here as to the question's answer, we surmise that key distinctions between the NHL and the NFL are the ages at which players are drafted, the ages at which latent talent becomes evident to evaluators, and the extent to which pre-professional leagues are concentrated (as in the NFL) or diverse (NHL).

In the NHL, 18 year olds can be drafted. Many drafted hockey players go on to play in college or in junior leagues before shifting to the NHL, if they transition to this league at all. In contrast, the NFL's rules on drafting specify that players must be at least three years removed from high school in order to be drafted. In almost all cases, NFL players drafted are at least 21 years of age. It is also worth pointing out that six months of an 18 year old's life is greater percentage-wise than six months of a 21 year old's life. In other words, all things equal, we would expect relative age effects to be stronger in an early-draft league like the NHL than in the NFL. To the extent that player talent evaluators are prone, mistakenly, to taking age into consideration when assessing a young player's potential, such a bias will be most pronounced in a sport where the minimum age for being drafted is relatively low. This would suggest a more general principle, namely, that sports with lower age limits for professional participation have greater age-related selection bias in their drafts. Addressing this conjecture will require research that spans multiple sports.

With respect to the point about the concentration of pre-professional play in football, we note that essentially all NFL players played several years of football in American colleges. This means that football draft evaluators have reliable and comparable information about

the talents of draftable players. In contrast, hockey draft evaluators are forced to draw comparisons between young players who are still in high school, players in American colleges, and players in Canadian junior professional leagues. Non-North American players further complicate the task of comparing the talent levels of potential professional hockey players.

We conclude with two broad points, one about the practical implications of our findings and a second on research design. With respect to the former, a practical implication of our results is that football talent evaluators should recognize that relatively older players within a given year cohort are better bets, all things equal, to be successful NFL players. Put another way, when NFL teams plan their draft strategies, they should take relative age into account. From the perspective of players, our findings imply that one's school district, and potentially one's position, are not negligible factors in determining success. Still, given the many thousands of high school and younger football players in the United States, the potential returns to strategic school district choice are microscopic in light of the ratio between the total number of affected births annually among professional football players and the set of potential professional football players.

With respect to research design, the approach that we have taken in our study of relative age in professional American football is based on monthly tables of births and goodness-of-fit statistics. This reflects the extant literature on age effects. However, this design has its limitations, particularly when considering a sport such as football. As we have discussed it here, football is potentially subject to a variety of age-related influences: calendar year effects beginning in January; school-related effects that are often fall-based; and, potential age effects induced by the fact that football is a fall-winter sport.

The combination of these effects means that, in the presence of relative age effects in football, we should not expect to see monotonically-decreasing birth counts from January to December. Put another way, rejecting that a set of observed football player births is drawn from a null distribution characterized by the monthly distribution of births across the United States does not necessarily identify what sort of an age effect, if any, is at work.

We can imagine two approaches that could address this feature of football. One, scholars could construct a parametric model of how a variety of age effects interact with each other and use this model to assess goodness of fit with an observed distribution of births. This seems to us to be a difficult task as it would require a researcher to have prior knowledge as to the strength of, say, calendar year age effects versus school cutoff age effects.

A second approach would be to rely on players whose birth months and life histories can identify the presence of a particular form of age effects. In principle, brothers who are professional football players yet have different birth months might be useful in this way – although we suspect that there are not enough siblings in the NFL to use familial connections as part of an age effects identification strategy. Notwithstanding this point, we have made progress in the sense of choosing specific players on which to focus by, in some of our analyses, limiting attention to football players who attended high school in states with fall cutoff dates.

More refined work in this direction could be done. With data including both athletes and non-athletes, one could focus attention solely on players born on opposite sides of a school cutoff. This cutoff should induce a discontinuity in the probability of becoming a professional football player. Similarly, individuals born the week before New Year's versus those born the week after should, in the presence of calendar year age effects, have different probabilities of becoming professional athletes. Moreover, individuals who move from one state to another, after starting high school, can shed light on school cutoff age effects if such moves are from, say, fall cutoff states to non-fall cutoff states. Lastly, and to the extent that football leagues in different countries are active at different times of the year, it might be possible to use players who move across national borders to estimate relative age effects induced by the timing of a particular sport.

The approaches sketched above transcend chi-squared goodness of fit tests that operate on monthly tables. As the statistical literature on age effects across various domains of human achievement continues to develop, we would expect to see research designs moving away from the aggregate, monthly analyses that up to this point have been very useful but can be confounded by a variety of mechanisms.

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